

The Editor-in-Chief
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Dear Editor-in-Chief,

We respectfully submit our manuscript, “**Limits of Fibonacci Index-Gap Coding: Complete Stream Costs, Compression Bounds, and Structural Controls,**” for consideration as a Research Article in *Information Processing & Management*.

Scope fit. The manuscript addresses a general information-processing problem: how to evaluate an intermediate representation when every stream-level model, delimiter, selector, length field, and padding bit required for exact reconstruction must be counted. It develops a complete-cost framework for amortized positional representations and applies it to a fully specified lossless stream based on Zeckendorf index gaps. The work therefore concerns information representation, algorithmic processing, quantitative evaluation, and reproducible system design.

Principal results.

- **Exact accounting:** A serialized format and a necessary-and-sufficient gain inequality determine whether a complete positional stream is shorter than its input, without treating metadata as free. Here, complete cost is defined within the serialized stream boundary.
- **Formal and empirical validation:** The transform is proved bijective; its sufficient term bound is calibrated on Canterbury and Silesia; and a compiled implementation verifies every decoding round trip on complete corpora and the 100 MB enwik8 benchmark.
- **Structural controls:** Lucas indices, ordinary binary set-bit positions, zero-block bitmaps, and boundary-zero trimming separate number-system effects from positional sparsity and simpler zero structure.
- **Reproducibility:** The public package provides the format implementation, tests, configurations, result tables, figure-generation scripts, and exact seeds.

Evidence and interpretation. A full-corpus sweep over five block sizes places the best FISA ratios at 92.15% for Canterbury and 98.78% for Silesia, but only one and two files compress, respectively. Matched zero-bitmap and zero-trim controls are smaller, and established codecs win every paired file comparison. The manuscript treats this outcome as a limit of the Fibonacci construction, not as evidence for a competitive general-purpose codec.

Relevance to readers. The framework is applicable to information-processing methods that introduce shared dictionaries, codebooks, model parameters, block selectors, or other amortized side information. It offers researchers and practitioners a reproducible procedure for separating payload reduction from complete, decodable system cost.

Author declarations.

- **Originality and exclusive submission:** The manuscript is original, has not been published previously, and is not under consideration elsewhere.
- **Author approval:** All named authors have approved the manuscript, the authorship order, and the stated contributions.
- **Competing interests and funding:** The authors declare no competing interests. This research received no external funding.
- **Ethical compliance:** The study involves no human participants, animals, personal data, or clinical interventions; it uses public benchmark corpora, synthetic data, and offline computational experiments.

We believe that the manuscript’s emphasis on rigorous information representation, transparent evaluation, and reproducible computational evidence will be relevant to the readership of *Information Processing & Management*. Thank you for considering this submission.

Yours sincerely,

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